

Generative Artificial Intelligence Group Project

Team name: Skynet

Topic: Cybersecurity Awareness (Cybersecurity education at school)

Title: Empowering Schools with Cyberbots

Abstract: “Skynets project, “Empowering Schools with Cyberbots” aims to address the need for cybersecurity awareness among students through implementing interactive technology. Since the digital space is growing, the growth comes with increased digital threats. Thus, it is a responsibility to educate young individuals and ensure they hold the skills to recognize and counteract potential cyber risks. We are developing the Cyberbot chatbot to be an engaging and educational tool designed to test students' understanding of cybersecurity and improve on it. This will be done through interactive quizzes and scenario-based questions.

The chatbot starts with opening a conversation with students and asking questions ranging from basic online safety practices to more complex cybersecurity scenarios. The questions are designed to test students' knowledge, and specifically may contain topics like password security, phishing scams, social media privacy, and safe internet usage. Skynet aims to develop a familiar chat-based interface, which makes learning about cybersecurity feel natural and enjoyable instead of an overwhelming chord. The quiz is targeted at high school students, which ensures that the chatbot provides valuable information relevant to their level of understanding.

Additionally, the chatbot has a dynamic scoring system that assesses every student's response and provides them with real-time feedback. Following a quiz session, students' scores are determined by how accurately they answered questions, giving them a summary of how well they performed as well as tips on how to strengthen their knowledge on cybersecurity. Constructive and supportive, this feedback acknowledges students' correct responses while empowering them to find areas for improvement.

The main advantage of the chatbot is that it can replicate actual cyberthreats in a safe setting. The chatbot may ask students how they would react in various scenarios, such as getting a questionable email or seeing odd prompts on a webpage. By encouraging critical thinking and problem-solving skills, these scenario-based questions promote a proactive approach to cybersecurity. This useful feature of the chatbot makes the information more memorable and applicable by enabling pupils to use it at once.

Additionally, the structure of the chatbot facilitates progressive learning. A slightly different set of questions or scenarios are shown to students each time they interact with the chatbot, which keeps the experience interesting and reinforces knowledge through a variety of instances. Because of its flexibility, students who regularly engage with the bot can deepen their comprehension of cybersecurity principles and, in the end, develop a more sophisticated grasp of digital safety.

Overall, Skynet aims to empower students by making cybersecurity information interesting and approachable, in addition to educating them. We hope to create a generation that is knowledgeable and equipped to securely traverse the digital world by providing a platform that evaluates and improves their cybersecurity awareness. Our chatbot is a prime example of leveraging technology to enhance education, showing how these bots can be crucial in increasing awareness and giving students the tools they need to stay safe online. One person at a time, our tool can lay the groundwork for a safe and knowledgeable online experience as cyberthreats change.”

AI Tools Utilized: LangChain and Groq